

UPMANYU JHA

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EXPERIENCE

Machine Learning Engineer (LLMs, NLP, CV) | *Quantiphi Analytics Solutions Private Limited* | Mumbai **Jul 2023 - Present**

- Designed, built, and deployed **15+** end-to-end **generative modeling pipeline** on Google Cloud (Vertex-AI, Document-AI), automating data extraction and classification for key insurance workflows
- Eliminated model **hallucinations** by **85%**, cut **GCP Cost** by **15%** and gained a **92%** average F1-score for critical fields from structured data from diverse unstructured documents for reducing manual processing time by over **40%**
- Implemented **generative multi-modal** model that **selectively verifies target signatures** on complex documents, bypassing extraneous parties which increased fraud detection recall by **35%** (maintaining precision) & cut the manual review queue by **50%**
- Led the end-to-end delivery of **4 major Full-Stack** Document-AI solutions, processing **70k+ documents** quarterly. Drove the optimization and refactoring of the core platform, boosting data extraction F1-score by **12%** and ensuring uptime
- Engineered a novel Retrieval-Augmented Generation (**RAG**) pipeline with fine-tuned LLMs, which elevated the F1-score for financial document extraction from **0.55 baseline to 0.90 (+0.35)**
- Championed a team-wide shift to **MLOps** best practices, establishing a **CI/CD pipeline** with automated testing raising **unit test coverage to 95%**. This initiative cut **production bugs** by **80%**, decreased deployment failures by **50%**
- Mentored **5 ML engineers** and **30 Business Analysts**, fostering a culture of high-quality software engineering across the team
- Resolved team's most critical Natural Language Processing (NLP) project blockers on GCP base DocAI platform, unblocking **10+ major issues** within 3 months and accelerating project delivery by **25%**

Security Researcher | Bug Bounty Hunter (Web Apps, API) | *Bugcrowd | Hackerone* | Remote **Jun 2021 - Jul 2023**

- Identified and reported **30+ critical vulnerabilities** in large-scale systems (including Amazon, Uber, Snapchat) by performing deep system analysis and threat modeling, demonstrating Full-Stack understanding of system architecture, **preventing** potential data breaches **affecting over 7M users** and **mitigating** an estimated **~\$1M+ in business risk**
- Pioneered **custom automation** tools in Python with **algorithm optimizations** to systematically test for system flaws and validate vulnerabilities, improving manual testing **efficiency by 60%** and enabling faster identification of architectural gaps and potential security biases, using Docker and Kubernetes on cloud infrastructure

AI & ENGINEERING PROJECTS / RESPONSIBILITY

Smart Traffic Management System | GANs, Reinforcement Learning (RL) and Computer Vision (CV) **Jan 2026 - Present**

- Designing an Ai-agent based traffic control prototype integrating YOLO-based video detection and multi-agent RL in a SUMO simulation to optimize urban signal timings
- Targeting **~10% commute time reduction** in average commute time across simulated intersections via real-time data streaming, state aggregation, and RL training pipelines using Python, PyTorch, OpenCV, Cursor-Ai and SUMO-RL

Image-Based DocuSearch | Python, CV, Tesseract OCR, NLTK, Scikit-learn, Elasticsearch, FastAPI **Oct 2025 - Dec 2025**

- Built an OCR search engine to **index 100,000+** scanned documents, boosting search relevance by **+0.20 nDCG@10** with a custom NLP pipeline to cleanse noisy OCR data
- Achieved **p95 query latency under 150ms** using an Elasticsearch and BM25 architecture

Teaching Assistant | Mentor - HAWK-i Labs **Aug 2021 - Present**

- Mentored 1000+ students in Computer Science, leading workshops on Python, Machine Learning (ML), Computer Vision and NLP

EDUCATION

B.E. in Computer Engineering | FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING **Graduated | May 2023**

- Key Coursework:** Machine Learning, Deep Learning, Neural Networks, Computer Vision, Recommendation System, Data Structure & Algorithm

TECHNICAL SKILLS

- Machine Learning:** PyTorch, TensorFlow, Keras, Hugging Face (Transformers), TensorRT, OpenCV, Scikit-Learn
- Artificial Intelligence:** NLP (Prompt Engineering, NLTK), Generative Models (GANs, RAG), Transfer Learning (LoRA), Model Context Protocol, Object Detection (YOLO)
- Data Visualization:** Matplotlib, Seaborn, Plotly, **Data Manipulation & Engineering:** Pandas, NumPy, OpenCV
- ML Optimization:** Distributed Training, Inference, Model compressing, Batch Processing
- Cloud Platforms & Database:** GCP (Doc AI, ADK, Vertex AI, GKE, Storage, BigQuery), AWS (SageMaker, S3, EC2, RDS)
- LLM Frameworks:** LangChain, crewAi, **Developer Tools & Frameworks:** Docker, Kubernetes, Github, Flask, Django, Postman
- Programming Languages:** Python, Go, SQL (PostgreSQL), Javascript, C++, C, Git, Bash Script, **Ticketing System:** Jira, Rally

CERTIFICATIONS AND TRAINING

Google Cloud Professional Cloud Security Engineer (GCP PCSE) Google Cloud	Dec 2025
Google Generative AI Leader (GCP Gen Ai) Google Cloud	Oct 2025
Certified Google Digital Lead (GCP CDL) Google Cloud	Sep 2025
Certified Google Cloud Professional Cloud Architect (GCP PCA) Google Cloud	Jun 2025
Certified Google Cloud Professional Machine Learning Engineer (GCP PMLE) Google Cloud	May 2024
Certified Google Cloud Associate Cloud Engineer (GCP ACE) Google Cloud	Dec 2023